

MATH 2123 - MATHEMATICS III

2024 Exam Paper: Beginner Guide + Full Solutions

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Designed for a student who is NOT yet an expert.

Each set includes: Topic -> Basics -> Problem-solving procedure -> Full solution -> Exam tips.

How to use this guide

1. Read the Basics first. Do not try to memorize the solution line-by-line.
2. Learn the Procedure as a reusable recipe. Most exam questions are variations of the same recipe.
3. Then follow the full worked solution and check the final answer.
4. Before the exam, revise the Final Answer boxes and the rapid revision pages at the end.

Paper map - what the 8 sets are really testing

Set	Main topic
1	Vector calculus: gradient, heat flow, curl/irrotational field, scalar potential, orthogonal surfaces
2	Vector integration: line integral/work, surface flux, vector-valued triple integral
3	Laplace transform: periodic functions, parameter/differentiation property, $\sin(at)/t$ and existence
4	Inverse Laplace: convolution, completing square, solving ODE by Laplace transform
5	Matrix basics: orthogonal/elementary/nonsingular, linear dependence, inverse by row operations
6	Rank and reduction: echelon/canonical/normal form, inverse, parameter-dependent linear system
7	Homogeneous systems and eigenvalues/eigenvectors/eigenspaces
8	Space curves: T-N-B frame, curvature, acceleration components, trihedral plane; Newton's law

A useful way to study this paper is to group Sets 1-4 as vector calculus + Laplace, and Sets 5-8 as matrices/eigen systems + curve mechanics.

SET 1 - Gradient, Conservative Fields and Orthogonal Surfaces

What this set tests

- 1(a) Gradient of a scalar field and the physical direction of heat flow.
- 1(b) Conditions for a vector field to be irrotational and how to recover a scalar potential.
- 1(c) Orthogonality of two surfaces using their normal vectors (gradients).

Basics you need

- For a scalar field $f(x,y,z)$, $\text{grad } f = (df/dx)i + (df/dy)j + (df/dz)k$. The gradient points toward the fastest increase of f .
- For temperature T , the heat-flow vector follows Fourier's idea $q = -k \text{ grad } T$. Therefore heat flows opposite to $\text{grad } T$ (for $k > 0$).
- For $F = P i + Q j + R k$, the field is irrotational when $\text{curl } F = 0$. Equivalently: $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$.
- If F is conservative, there is a scalar potential ϕ such that $\text{grad } \phi = F$.
- A surface $f=0$ has normal vector $\text{grad } f$. Two surfaces are orthogonal at a point when $\text{grad } f \cdot \text{grad } g = 0$ at that point.

Reusable procedure

Gradient question: differentiate $\rightarrow$ substitute point $\rightarrow$ compute magnitude/direction.

Irrotational question: compare cross-partial derivatives $\rightarrow$ find constants $\rightarrow$ integrate one component $\rightarrow$ match the others.

Orthogonal surfaces: first use the fact that the given point lies on the surface; then dot the two gradients and set the result to zero.

1(a) Temperature gradient and direction of maximum heat flow

Given: $T(x,y,z) = 40 - 0.1x^2 - 0.05y^2 - 0.2z$.

$$\text{grad } T = (-0.2x)i + (-0.1y)j + (-0.2)k$$

At $(2,3,0.5)$:

$$\text{grad } T = -0.4 i - 0.3 j - 0.2 k$$

$$|\text{grad } T| = \sqrt{0.4^2 + 0.3^2 + 0.2^2} = \sqrt{0.29} = \sqrt{29}/10$$

The unit vector of maximum temperature increase is:

$$u_{\text{increase}} = \text{grad } T / |\text{grad } T| = (-4 i - 3 j - 2 k)/\sqrt{29}$$

But heat flows from hotter to colder regions, so the direction of maximum heat flow is the opposite direction:

$$u_{\text{heat}} = -\text{grad } T / |\text{grad } T| = (4 i + 3 j + 2 k)/\sqrt{29}$$

Final answer

$\text{grad } T(2,3,0.5) = -0.4i - 0.3j - 0.2k$; maximum heat-flow direction $= (4i+3j+2k)/\sqrt{29}$.

Common mistake / exam note

Some books loosely say "direction of maximum heat flow" when they mean maximum increase of temperature. In physics, actual heat flow is opposite $\text{grad } T$. State both directions clearly if the wording is ambiguous.

1(b) Make F irrotational and find its scalar potential

$$F = (x + 2y + az)i + (bx - 3y - z)j + (4x + cy + 2z)k.$$

Let $P=x+2y+az$, $Q=bx-3y-z$, $R=4x+cy+2z$. For an irrotational field:

$$P_y = Q_x \rightarrow 2 = b \rightarrow b = 2$$

$$P_z = R_x \rightarrow a = 4$$

$$Q_z = R_y \rightarrow -1 = c \rightarrow c = -1$$

Now $F = (x+2y+4z)i + (2x-3y-z)j + (4x-y+2z)k$. Let $\text{grad } \phi = F$.

Integrate $\phi_x = x+2y+4z$ with respect to x :

$$\phi = x^2/2 + 2xy + 4xz + f(y, z)$$

Differentiate with respect to y and compare with Q :

$$\phi_y = 2x + f_y = 2x - 3y - z \rightarrow f_y = -3y - z$$

$$f(y, z) = -3y^2/2 - yz + g(z)$$

Differentiate with respect to z and compare with R :

$$\phi_z = 4x - y + g'(z) = 4x - y + 2z \rightarrow g'(z) = 2z \rightarrow g = z^2 + C$$

Final answer

$$a=4, b=2, c=-1; \phi = x^2/2 + 2xy + 4xz - 3y^2/2 - yz + z^2 + C.$$

1(c) Find a and b so two surfaces are orthogonal

Surface 1: $ax^2 - byz = (a+2)x$. Write $f = ax^2 - byz - (a+2)x = 0$.

Surface 2: $4x^2y + z^3 = 4$. Write $g = 4x^2y + z^3 - 4 = 0$.

Step 1: the point $(1, -1, 2)$ lies on surface 1:

$$a(1)^2 - b(-1)(2) = (a+2)(1) \rightarrow a + 2b = a + 2 \rightarrow b=1$$

Step 2: compute the normals.

$$\text{grad } f = (2ax - (a+2), -bz, -by)$$

$$\text{grad } f \text{ at } (1, -1, 2) = (a-2, -2b, b)$$

$$\text{grad } g = (8xy, 4x^2, 3z^2) \rightarrow \text{grad } g \text{ at } (1, -1, 2) = (-8, 4, 12)$$

Orthogonality means dot product = 0:

$$-8(a-2) + 4(-2b) + 12b = 0$$

$$-8a + 16 + 4b = 0 \rightarrow b = 2a - 4$$

Using $b=1$: $1 = 2a - 4 \rightarrow a = 5/2$.

Final answer

$$a = 5/2 \text{ and } b = 1.$$

SET 2 - Line, Surface and Volume Integrals

What this set tests

- 2(a) Work done by a force field along two different paths.
- 2(b) Flux through a triangular portion of a plane.
- 2(c) Triple integration of a vector field over a bounded volume.

Basics you need

- Work along curve C: $W = \int_C \mathbf{F} \cdot d\mathbf{r}$. Parametrize the curve, compute $d\mathbf{r}$, substitute into $\mathbf{F}$, and integrate.
- Flux through a graph $z=z(x,y)$: $\mathbf{F} \cdot \mathbf{n} \, dS$ can be written $\mathbf{F} \cdot (-z_x, -z_y, 1) \, dx \, dy$ for the upward orientation.
- A vector-valued triple integral is done component by component.

Reusable procedure

Line integral: choose ONE parameter $\rightarrow$ write $x,y,z \rightarrow$ differentiate $\rightarrow$ form $\mathbf{F} \cdot \mathbf{r}'(\text{parameter}) \rightarrow$ integrate.

Surface flux: solve the surface for one variable $\rightarrow$ choose orientation $\rightarrow$ find projected region $\rightarrow$ integrate.

Triple integral: sketch/understand the bounding surfaces $\rightarrow$ choose limits from inside to outside $\rightarrow$ integrate each component.

2(a) Work done along two paths

$$\mathbf{F} = 3x^2 \mathbf{i} + (2xz-y)\mathbf{j} + z \mathbf{k}.$$

(i) Straight line from (0,0,0) to (2,1,3)

Use $\mathbf{r}(t) = (2t, t, 3t)$, $0 \leq t \leq 1$. Then $\mathbf{r}'(t) = (2, 1, 3)$.

$$\mathbf{F}(\mathbf{r}(t)) = (12t^2, 12t^2 - t, 3t)$$

$$\mathbf{F} \cdot \mathbf{r}' = 12t^2(2) + (12t^2 - t)(1) + 3t(3) = 36t^2 + 8t$$

$$W = \int_0^1 (36t^2 + 8t) \, dt = 12 + 4 = 16$$

Final answer

Work along the straight line = 16.

(ii) Curve $x^2=4y$, $3x^3=8z$, from $x=0$ to $x=2$

Use x itself as the parameter: $y=x^2/4$ and $z=3x^3/8$.

$$d\mathbf{r} = (1, x/2, 9x^2/8) \, dx$$

$$\mathbf{F} = (3x^2, (3x^4 - x^2)/4, 3x^3/8)$$

$$\begin{aligned} \mathbf{F} \cdot d\mathbf{r} &= 3x^2 + [(3x^4 - x^2)/4](x/2) + (3x^3/8)(9x^2/8) \\ &= 3x^2 - x^3/8 + 51x^5/64 \end{aligned}$$

$$W = \int_0^2 [3x^2 - x^3/8 + 51x^5/64] \, dx = 8 - 1/2 + 17/2 = 16$$

Final answer

Work along the curved path = 16.

Common mistake / exam note

The two answers happen to be equal, but do not automatically conclude the whole field is conservative. A proper conservative-field test would check $\text{curl } \mathbf{F} = 0$ in the region.

2(b) Flux through $x+y+2z=4$ in the first octant

$\mathbf{F} = (2x^2-3z)\mathbf{i} - 2xy\mathbf{j} - 4x\mathbf{k}$. The plane gives $z=(4-x-y)/2$.

For the normal pointing away from the origin (the outward normal of the tetrahedron), use:

$$(-z_x, -z_y, 1) = (1/2, 1/2, 1)$$

The projection D in the xy-plane is the triangle $x \geq 0, y \geq 0, x+y \leq 4$.

$$\mathbf{F} \cdot d\mathbf{S} = [(2x^2-3z)/2 - xy - 4x] dx dy$$

Substitute $z=(4-x-y)/2$:

$$\mathbf{F} \cdot d\mathbf{S} = [x^2 - xy - 13x/4 + 3y/4 - 3] dx dy$$

$$\text{Flux} = \int_{x=0}^4 \int_{y=0}^{4-x} (\dots) dy dx = -40$$

Final answer

Flux = -40 for the outward/away-from-origin orientation. With the opposite orientation, the flux is +40.

2(c) Vector triple integral

$\mathbf{F} = 2xz\mathbf{i} - x\mathbf{j} + y^2\mathbf{k}$. The region is bounded by $x=0, y=0, y=6, z=x^2, z=9$.

Because $z=x^2$ meets $z=9$ at $x=3$, convenient limits are:

$$0 \leq x \leq 3, \quad 0 \leq y \leq 6, \quad x^2 \leq z \leq 9$$

Integrate each component separately.

$$\text{i-component} = \int_0^3 \int_0^6 \int_{x^2}^9 2xz dz dy dx = 1458$$

$$\text{j-component} = \int_0^3 \int_0^6 \int_{x^2}^9 (-x) dz dy dx = -243/2$$

$$\text{k-component} = \int_0^3 \int_0^6 \int_{x^2}^9 y^2 dz dy dx = 1296$$

Final answer

Triple integral of $\mathbf{F}$ over $V = 1458\mathbf{i} - (243/2)\mathbf{j} + 1296\mathbf{k}$.

SET 3 - Laplace Transform: Periodic Functions and Special Properties

What this set tests

- 3(a) Laplace transform of a periodic piecewise function.
- 3(b) Turning an improper integral into a Laplace-transform calculation.
- 3(c) Transform of $\sin(at)/t$ and the existence test for $\cos(at)/t$.

Basics you need

- If $f(t)$ has period T , then $L\{f\} = [\int_0^T e^{-st} f(t) dt] / [1 - e^{-sT}]$.
- Differentiation property: if $F(s) = L\{f(t)\}$, then $L\{t^n f(t)\} = (-1)^n d^n F/ds^n$.
- A Laplace transform must converge near $t=0$ and as $t \rightarrow \infty$. A $1/t$ singularity at $t=0$ usually causes trouble unless the numerator also goes to zero.

3(a) Periodic piecewise function

$F(t)=3t$ for $0 < t < 2$; $F(t)=6$ for $2 < t < 4$; and $F(t+4)=F(t)$. Period $T=4$.

$$L\{F\} = 1/(1 - e^{-4s}) \left[\int_0^2 3t e^{-st} dt + \int_2^4 6e^{-st} dt \right]$$

Evaluate the first part by integration by parts and the second directly:

$$\int_0^2 3t e^{-st} dt = 3[1 - e^{-2s}](1+2s)/s^2$$

$$\int_2^4 6e^{-st} dt = 6(e^{-2s} - e^{-4s})/s$$

After combining and simplifying:

$$L\{F\} = 3[1 - e^{-2s} - 2s e^{-4s}] / [s^2(1 - e^{-4s})]$$

Final answer

$$L\{F(t)\} = 3(1 - e^{-2s} - 2s e^{-4s}) / \{s^2(1 - e^{-4s})\}.$$

3(b) Evaluate $\int_0^\infty t^3 e^{-t} \sin t dt$

Start from $L\{\sin t\} = 1/(s^2+1)$.

$$\begin{aligned} L\{t^3 \sin t\} &= (-1)^3 d^3/ds^3 [1/(s^2+1)] \\ &= 24s(s^2-1)/(s^2+1)^4 \end{aligned}$$

The extra factor e^{-t} means evaluate the Laplace transform at $s=1$:

$$\int_0^\infty t^3 e^{-t} \sin t dt = 24(1)(1^2-1)/(1^2+1)^4 = 0$$

Final answer

The improper integral equals 0.

3(c) $L\{\sin(at)/t\}$; does $L\{\cos(at)/t\}$ exist?

Let $I(a) = \int_0^\infty e^{-st} \sin(at)/t \, dt$. Differentiate with respect to a :

$$dI/da = \int_0^\infty e^{-st} \cos(at) \, dt = s/(s^2 + a^2)$$

Since $I(0) = 0$:

$$I(a) = \int_0^a s/(s^2 + u^2) \, du = \tan^{-1}(a/s), \quad s > 0$$

Final answer

$L\{\sin(at)/t\} = \tan^{-1}(a/s)$, for $s > 0$.

For $\cos(at)/t$, as $t \rightarrow 0$ we have $\cos(at) \sim 1$ and $e^{-st} \sim 1$, so the integrand behaves like $1/t$. The integral $\int_0^\epsilon dt/t$ diverges.

Final answer

The ordinary Laplace transform of $\cos(at)/t$ does NOT exist because of the non-integrable singularity at $t=0$.

SET 4 - Inverse Laplace, Convolution and ODE by Laplace

What this set tests

- 4(a) Convolution theorem for an inverse transform.
- 4(b) Completing the square and using the shift theorem.
- 4(c) Solving a second-order differential equation by Laplace transform when one condition is given at $t=\pi/2$.

Basics you need

- Convolution: if $F(s)=L\{f\}$ and $G(s)=L\{g\}$, then $L^{-1}\{F(s)G(s)\} = (f*g)(t) = \int_0^t f(u)g(t-u)du$.
- Shift rule: $L^{-1}\{(s-a)/[(s-a)^2+b^2]\} = e^{at} \cos bt$ and $L^{-1}\{b/[(s-a)^2+b^2]\} = e^{at} \sin bt$.
- For y'' , $L\{y''\}=s^2Y-sy(0)-y'(0)$. If $y'(0)$ is unknown, keep it as a constant and use the later boundary condition to determine it.

4(a) Convolution theorem

Find $L^{-1}\{1/[(s+1)(s^2+1)]\}$. Choose:

$$1/(s+1) \leftrightarrow e^{-t}, \quad 1/(s^2+1) \leftrightarrow \sin t$$

$$f*g = \int_0^t e^{-u} \sin(t-u) du$$

Put $v=t-u$ or integrate directly:

$$\int_0^t e^{-u} \sin(t-u) du = [\sin t - \cos t + e^{-t}]/2$$

Final answer

$$L^{-1}\{1/[(s+1)(s^2+1)]\} = (\sin t - \cos t + e^{-t})/2.$$

4(b) Inverse transform of $s/(s^2-2s+35)$

$$s^2 - 2s + 35 = (s-1)^2 + 34$$

$$s = (s-1) + 1$$

$$s/[(s-1)^2+34] = (s-1)/[(s-1)^2+34] + 1/[(s-1)^2+34]$$

Let $b=\sqrt{34}$. The second numerator 1 is $(1/b)*b$.

Final answer

$$L^{-1}\{s/(s^2-2s+35)\} = e^t \cos(\sqrt{34}t) + [e^t/\sqrt{34}] \sin(\sqrt{34}t).$$

4(c) Solve $y'' + 9y = \cos 2t$, $y(0)=1$, $y(\pi/2)=-1$

Let $c=y'(0)$, initially unknown. Taking Laplace transforms:

$$(s^2 Y - s - c) + 9Y = s/(s^2+4)$$

$$Y = (s+c)/(s^2+9) + s/[(s^2+4)(s^2+9)]$$

Use partial fractions on the last term:

$$s/[(s^2+4)(s^2+9)] = (1/5)[s/(s^2+4) - s/(s^2+9)]$$

Invert:

$$\begin{aligned} y(t) &= \cos 3t + (c/3)\sin 3t + (1/5)(\cos 2t - \cos 3t) \\ &= (4/5)\cos 3t + (c/3)\sin 3t + (1/5)\cos 2t \end{aligned}$$

Apply $y(\pi/2)=-1$: $\cos(3\pi/2)=0$, $\sin(3\pi/2)=-1$, $\cos(\pi)=-1$.

$$-c/3 - 1/5 = -1 \quad \rightarrow \quad c = 12/5$$

Final answer

$$y(t) = (4/5)\cos 3t + (4/5)\sin 3t + (1/5)\cos 2t.$$

SET 5 - Matrix Types, Linear Dependence and Matrix Inverse

What this set tests

- 5(a) Definitions, examples and properties of common matrix classes.
- 5(b) Linear dependence among vectors.
- 5(c) Gauss-Jordan method for a matrix inverse.

5(a) Definitions, examples and two useful properties

(i) Orthogonal matrix

Definition: A square real matrix Q is orthogonal if $Q^T Q = I$ (equivalently $Q^{-1}=Q^T$).

Example: $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

- Its rows and columns are orthonormal vectors.
- $\det(Q)=+1$ or -1 , so every orthogonal matrix is nonsingular.

(ii) Elementary matrix

Definition: An elementary matrix is obtained by performing exactly one elementary row operation on an identity matrix.

Example: $E = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ (obtained from I by $R_2 \rightarrow R_2 + 3R_1$)

- Every elementary matrix is nonsingular and its inverse is also elementary.
- Left multiplication EA performs on A the same row operation used to create E from I .

(iii) Nonsingular matrix

Definition: A square matrix A is nonsingular (invertible) if A^{-1} exists; equivalently $\det(A) \neq 0$.

Example: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\det = -2 \neq 0$

- $\text{rank}(A)=n$ for an $n \times n$ nonsingular matrix.
- $Ax=0$ has only the trivial solution $x=0$.

5(b) Test linear independence

$v_1=(0,2,3,4)$, $v_2=(2,3,0,1)$, $v_3=(4,8,3,6)$.

Try to express v_3 as $\alpha v_1 + \beta v_2$. From the first component: $2\beta=4 \rightarrow \beta=2$. From the second: $2\alpha+3(2)=8 \rightarrow \alpha=1$.

Check the remaining components: $3(1)+0(2)=3$ and $4(1)+1(2)=6$. Both match.

$$v_3 = v_1 + 2v_2$$

Final answer

The three vectors are linearly dependent. One dependence relation is $v_3 - v_1 - 2v_2 = 0$.

Common mistake / exam note

The wording "express an independent vector as a linear combination of dependent vectors" is unusual. The standard task is to express one dependent vector as a linear combination of the other vectors, as done above.

5(c) Inverse by elementary row transformations

$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$. Start with $[A \mid I]$.

$$\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 0 & 1 \end{array}$$

Use these row operations:

1. $R1 \leftrightarrow R2$
2. $R3 \rightarrow R3 - 3R1$
3. $R3 \rightarrow R3 + 5R2$
4. $R3 \rightarrow (1/2)R3$
5. $R2 \rightarrow R2 - 2R3$
6. $R1 \rightarrow R1 - 3R3$
7. $R1 \rightarrow R1 - 2R2$

The left side becomes I and the right side becomes A^{-1} :

$$A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$$

Final answer

A is invertible ($\det A = -2$) and its inverse is the matrix shown above.

SET 6 - Rank, Echelon/Canonical/Normal Form and Parameter Systems

What this set tests

- 6(a) Rank of a matrix, systematic row reduction, inverse as a by-product.
- 6(b) Classifying a linear system as unique / infinitely many / no solution as a parameter K changes.

Basics you need

- Rank = number of pivots (nonzero rows in row-echelon form).
- Row-echelon form has zeros below each pivot. Reduced/canonical form additionally has each pivot equal to 1 and zeros above it.
- Normal form under row/column operations is $\text{diag}(I_r, 0)$. For a full-rank 3×3 matrix this is simply I_3 .
- For $Ax=b$: unique if $\text{rank } A = \text{rank}[A|b] = \text{number of unknowns}$; infinitely many if equal ranks are smaller; no solution if the two ranks differ.

6(a) Reduce A, find rank and inverse

$$A = \begin{bmatrix} 0 & 3 & -1 \\ 1 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}.$$

Start row reduction:

$$\begin{bmatrix} 0 & 3 & -1 \\ 1 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

8. $R_1 \leftrightarrow R_2$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & -1 \\ -2 & 2 & 1 \end{bmatrix}$$

9. $R_3 \rightarrow R_3 + 2R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 4 & 5 \end{bmatrix}$$

10. $R_3 \rightarrow R_3 - (4/3)R_2$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 19/3 \end{bmatrix}$$

This is an echelon form. It has 3 nonzero rows, so $\text{rank}(A)=3$.

Continuing Gauss-Jordan reduction gives the canonical/reduced form I_3 . Since $\text{rank}=3$, the normal form is also I_3 .

Final answer

$\text{rank}(A)=3$; echelon form can be $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 19/3 \end{bmatrix}$; canonical and normal forms are I_3 .

Inverse as a by-product

Apply the same row operations to $[A|I]$. A convenient continuation is: swap R_1, R_2 ; $R_3 \rightarrow R_3 + 2R_1$; then $R_3 \rightarrow 3R_3 - 4R_2$; scale R_3 by $1/19$; eliminate the third and second columns.

$$A^{-1} = (1/19) \begin{bmatrix} 3 & 5 & -7 \\ 5 & 2 & 1 \\ -4 & 6 & 3 \end{bmatrix}$$

Final answer

$$A^{-1} = \begin{bmatrix} 3/19 & 5/19 & -7/19 \\ 5/19 & 2/19 & 1/19 \\ -4/19 & 6/19 & 3/19 \end{bmatrix}.$$

6(b) Classify the K-dependent system

System matrix and right-hand side:

$$\begin{bmatrix} K & 3 & 2 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}^T = \begin{bmatrix} K \\ 1 \\ 1 \end{bmatrix}^T$$

Its determinant is:

$$\det A = 9(K-1)$$

Therefore for $K \neq 1$, $\det A \neq 0$ and the solution is unique. Solving the equations gives:

$$x=1, \quad y=2/9, \quad z=-1/3$$

For $K=1$, the first and third equations become identical. Then $\text{rank } A = \text{rank}[A|b] = 2 < 3$, so there is one free variable and infinitely many solutions.

There is no value of K that makes $\text{rank}[A|b]$ greater than $\text{rank } A$, so there is no inconsistent/no-solution case.

Final answer

Unique solution for $K \neq 1$: $(x,y,z)=(1,2/9,-1/3)$. Infinitely many solutions for $K=1$. No value of K gives no solution.

SET 7 - Homogeneous Systems and Eigensystems

What this set tests

7(a) When a homogeneous linear system has nontrivial solutions.

7(b) Eigenvalues, eigenvectors, eigenspaces, and multiplicities; effect of shifting a matrix by $4I$.

Basics you need

- A homogeneous square system $Ax=0$ always has the trivial solution. It has a nontrivial solution exactly when $\det(A)=0$.
- Eigenvalue equation: $Av=\lambda v$ with $v \neq 0$. Find λ from $\det(A-\lambda I)=0$, then solve $(A-\lambda I)v=0$.
- The eigenspace for λ is $\text{Null}(A-\lambda I)$.
- Algebraic multiplicity (AM) is how many times λ repeats as a root of the characteristic polynomial. Geometric multiplicity (GM) is the dimension of its eigenspace, and $1 \leq GM \leq AM$.

7(a) Find T for trivial/nontrivial solutions

Equations: $-2x+y+3z=0$, $2y+z=0$, $x-4y+Tz=0$.

$$A = \begin{bmatrix} -2 & 1 & 3 \\ 0 & 2 & 1 \\ 1 & -4 & T \end{bmatrix}$$

$$\det A = -4T - 13$$

If $\det A \neq 0$, the only solution is $x=y=z=0$. Therefore the system has only the trivial solution when $T \neq -13/4$.

For a nontrivial solution set $\det A=0$:

$$-4T - 13 = 0 \quad \rightarrow \quad T = -13/4$$

At $T=-13/4$, use $2y+z=0 \rightarrow z=-2y$. The first equation gives $-2x+y+3(-2y)=0 \rightarrow -2x-5y=0 \rightarrow x=-5y/2$.

Choose $y=-2c$ to remove fractions. Then $x=5c$ and $z=4c$.

Final answer

Only trivial solution for $T \neq -13/4$. Nontrivial solutions for $T=-13/4$: $(x,y,z)=c(5,-2,4)$, c arbitrary.

7(b) Eigensystem of $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Two useful general properties:

- Eigenvectors belonging to distinct eigenvalues are linearly independent.
- For each eigenvalue, geometric multiplicity $\leq$ algebraic multiplicity; a matrix is diagonalizable when it has enough independent eigenvectors (sum of GMs = n).

Characteristic equation:

$$\det(A - \lambda I) = (1 - \lambda)^2 - 1 = \lambda(\lambda - 2) = 0$$

Eigenvalues: $\lambda = 2$ and $\lambda = 0$

For $\lambda = 2$: $(A - 2I)v = 0$ gives $x = y$, so eigenspace $E_2 = \text{span}\{(1, 1)\}$.

For $\lambda = 0$: $Av = 0$ gives $x + y = 0$, so eigenspace $E_0 = \text{span}\{(1, -1)\}$.

Each eigenvalue appears once, so $AM = 1$ and $GM = 1$ for both.

Final answer

Eigenpairs may be written $(2, (1, 1))$ and $(0, (1, -1))$; $E_2 = \text{span}\{(1, 1)\}$, $E_0 = \text{span}\{(1, -1)\}$; $AM = GM = 1$ for each.

Eigenvalues/eigenvectors of $A - 4I$

Subtracting $4I$ shifts every eigenvalue by -4 but does not change the eigenvectors.

$$\lambda(A - 4I): 2 - 4 = -2, \quad 0 - 4 = -4$$

For -2 : eigenvectors $\text{span}\{(1, 1)\}$; for -4 : eigenvectors $\text{span}\{(1, -1)\}$

Final answer

$A - 4I$ has eigenvalues -2 and -4 with the same corresponding eigendirections $(1, 1)$ and $(1, -1)$.

SET 8 - Space Curves, T-N-B Frame, Curvature and Particle Motion

What this set tests

8(a) Unit tangent T , principal normal N , binormal B , curvature/radius, acceleration components, and a trihedral plane.

8(b) Newton's second law $F=ma$ and integration of acceleration to velocity.

Basics you need

- Velocity $v=r'(t)$; speed $=|r'(t)|$; unit tangent $T=r'/|r'|$.
- Principal normal $N = T'/|T'|$ (when T' is nonzero). Binormal $B=T \times N$.
- Curvature $\kappa = |T'|/|r'| = |r' \times r''|/|r'|^3$; radius of curvature $\rho=1/\kappa$.
- Acceleration decomposes as $a = a_T T + a_N N$ where $a_T=d(\text{speed})/dt$ and $a_N=\text{speed}^2 \kappa$.
- Trihedral planes at a point: osculating plane spans T, N ; normal plane spans N, B ; rectifying plane spans T, B .

8(a) Curve $r(t)=5 \cos t \mathbf{i} + 5 \sin t \mathbf{j} - t \mathbf{k}$

$$r'(t)=(-5 \sin t, 5 \cos t, 0), \quad |r'|=5$$

$$T(t)=(-\sin t, \cos t, 0)$$

$$T'(t)=(-\cos t, -\sin t, 0), \quad |T'|=1$$

$$N(t)=(-\cos t, -\sin t, 0)$$

$$B=T \times N=(0, 0, 1)=\mathbf{k}$$

Curvature and radius:

$$\kappa = |T'|/|r'| = 1/5, \quad \rho=1/\kappa=5$$

Acceleration:

$$a=r''(t)=(-5 \cos t, -5 \sin t, 0)=5N$$

$$a_T=d|v|/dt=0, \quad a_N=|v|^2 \kappa=25(1/5)=5$$

Final answer

$T=(-\sin t, \cos t, 0)$, $N=(-\cos t, -\sin t, 0)$, $B=(0, 0, 1)$, $\kappa=1/5$, $\rho=5$, $a_T=0$, $a_N=5$, and $a=5N$.

Plane containing B and N at $t=\pi/2$

At $t=\pi/2$, $r=(0, 5, -1)$, $T=(-1, 0, 0)$, $N=(0, -1, 0)$, $B=(0, 0, 1)$. The plane spanned by N and B is the normal plane; its normal vector is T .

$$T \cdot [(x, y, z) - (0, 5, -1)] = 0 \quad \rightarrow \quad -(x-0)=0$$

Final answer

The required normal (trihedral) plane is $x=0$.

For orientation: the other two trihedral planes at this point are the osculating plane $z=-1$ and the rectifying plane $y=5$.

8(b) Find velocity from force

$F = -t \mathbf{i} + t^2 \mathbf{j}$, mass $m=2$, and $\mathbf{v}(0)=5\mathbf{i}$.

$$\mathbf{a} = F/m = -(t/2)\mathbf{i} + (t^2/2)\mathbf{j}$$

Integrate acceleration:

$$\mathbf{v}(t) = (-t^2/4 + C_1)\mathbf{i} + (t^3/6 + C_2)\mathbf{j}$$

Use $\mathbf{v}(0)=5\mathbf{i}$, so $C_1=5$ and $C_2=0$.

Final answer

$$\mathbf{v}(t) = (5 - t^2/4)\mathbf{i} + (t^3/6)\mathbf{j}.$$

Rapid Revision - Procedures to Memorize

Topic	Recipe
Gradient / heat	$\text{grad } T = \text{partial derivatives}$. Maximum increase is $+\text{grad direction}$; physical heat flow is $-\text{grad direction}$.
Irrotational field	Set $\text{curl } F=0$ by equating cross partials. Then integrate P to get ϕ and use Q, R to determine missing functions.
Orthogonal surfaces	Use the point-on-surface condition first. Then $\text{grad } f \cdot \text{grad } g=0$.
Work integral	Parametrize $r(t)$, compute $r'(t)$, integrate $F(r(t)) \cdot r'(t)$.
Surface flux	Choose orientation, get vector area element, find projected region, integrate.
Periodic Laplace	$L\{f\} = \text{integral over one period} / (1 - e^{-sT})$.
Convolution	Product in s -domain becomes convolution integral in t -domain.
Inverse by rows	Augment $[A I]$ and row-reduce left side to I ; right side becomes A^{-1} .
Rank/system	Compare $\text{rank } A$ and $\text{rank}[A b]$. For square systems, $\det A \neq 0$ immediately means unique.
Nontrivial homogeneous	For square $Ax=0$, nontrivial solution iff $\det A=0$.
Eigenproblem	$\det(A - \lambda I)=0 \rightarrow \lambda$; then solve $\text{null}(A - \lambda I)$.
T-N-B curve	$T=r'/ r' $, $N=T'/ T' $, $B=T \times N$, $\kappa= T' / r' $.
Particle force	$a=F/m$; integrate a to get v ; use initial velocity to find constants.

Final answers at a glance

Question	Answer
1(a)	$\text{grad } T=(-0.4, -0.3, -0.2)$; heat direction $=(4, 3, 2)/\sqrt{29}$.
1(b)	$a=4, b=2, c=-1$; $\phi=x^2/2+2xy+4xz-3y^2/2-yz+z^2+C$.
1(c)	$a=5/2, b=1$.
2(a)	$W=16$ for both given paths.
2(b)	Flux $=-40$ for outward/away-from-origin normal.
2(c)	$1458i - (243/2)j + 1296k$.
3(a)	$3(1 - e^{-2s} - 2s e^{-4s})/[s^2(1 - e^{-4s})]$.
3(b)	0.
3(c)	$L\{\sin(at)/t\} = \tan^{-1}(a/s)$; $L\{\cos(at)/t\}$ does not exist ordinarily.
4(a)	$(\sin t - \cos t + e^{-t})/2$.

4(b)	$e^t \cos(\sqrt{3}t) + (e^t/\sqrt{3})\sin(\sqrt{3}t)$.
4(c)	$y = (4/5)\cos 3t + (4/5)\sin 3t + (1/5)\cos 2t$.
5(b)	Dependent: $v_3 = v_1 + 2v_2$.
5(c)	$A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & -1 \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$.
6(a)	$\text{rank}=3$; $A^{-1} = (1/19)\begin{bmatrix} 3 & 5 & -7 \\ 5 & 2 & 1 \\ -4 & 6 & 3 \end{bmatrix}$.
6(b)	$K \neq 1$ unique $(1, 2/9, -1/3)$; $K=1$ infinite; no no-solution K .
7(a)	$T = -13/4$ for nontrivial; solution $c(5, -2, 4)$.
7(b)	A : $\lambda = 2 \rightarrow \text{span}(1, 1)$, $\lambda = 0 \rightarrow \text{span}(1, -1)$; $A - 4I$: $-2, -4$ same eigenvectors.
8(a)	$T = (-\sin, \cos, 0)$, $N = (-\cos, -\sin, 0)$, $B = k$, $\kappa = 1/5, \rho = 5, a_T = 0, a_N = 5$; plane $x = 0$.
8(b)	$v = (5 - t^2/4)i + (t^3/6)j$.

Exam Strategy for This Paper

If you are weak in the course, prioritize in this order

1. Set 5 + Set 6: row operations, rank and inverse are highly procedural.
2. Set 3 + Set 4: Laplace questions follow repeatable formulas.
3. Set 7: determinant + eigenvalue procedure is also systematic.
4. Set 1 + Set 2: vector calculus needs more interpretation of geometry and direction.
5. Set 8: learn the T-N-B formulas and one clean worked example.

In a 3-hour exam, write the formula first, define your parameter/limits clearly, and keep one line per logical step. This reduces sign mistakes and also earns method marks even if the final arithmetic goes wrong.

One final habit

After every solution, spend 10 seconds checking: dimensions/shape of the answer, sign/orientation, limits, and whether all parts of the question were answered.